



Nirma University
Institute of Technology
B. Tech in Electronics and Communication
Engineering
Semester – V(ODD 2023)
2EC501 – VLSI Design

SPECIAL ASSIGNMENT
21BEC005
4:1 MULTIPLEXER

Table of Contents:

- 1)Introduction to Multiplexer
- 2)4:1 Multiplexer
- 2.1)Truth table
- 2.2)Expanded truth table
- 3)Transistors
- 3.1)NMOS transistor
- 3.2)PMOS transistor
- 3.3)CMOS transistor
- 4)Question and Answers
- 4.1)Optimized boolean equation
- 4.2)Optimized gate level circuit diagram
- 4.3)Transistor level schematic for CMOS implementation
- 4.4) Transistor level schematic using NMOS pass gate
- 4.5) Transistor level schematic using CMOS pass gate
- 4.6) Stick Diagram for 4:1 MUX
- 4.7) DSCH implementation for 4:1 MUX
- 4.8)Layout for 4:1 MUX using Microwind.
- 4.9)Simulation of layout and observations
- 5)Conclusion

1)Introduction to multiplexer

Multiplexer is a combinational circuit. It has 2^n inputs line and one output line. In other word the multiplexer is a multiple inputs and one output. The binary information is place on the input line and directed to output line. The output line selection is depends upon on the select line,Unlike encoder and decoder which uses n inputs lines and m input lines. The multiplexer is also treated as a MUX. Generally the number of input lines depends on the power of two such as 2, 4, 8, 16, etc. some of the most commonly usable multiplexer is 2 to 1, 4 to 1, 8 to 1 and 16 to 1 multiplexers. Multiplexer is also available in IC forms various input and select line configurations. Some multiplexer ICs are listed below.

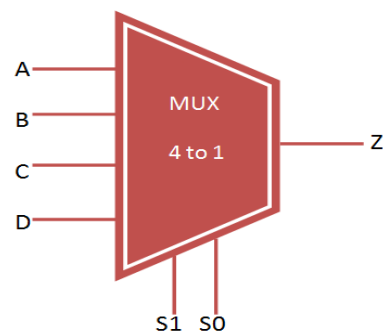
1. 74157 = Quad 2-to-1 multiplexer
2. 8158 = Quad 2-to-1 multiplexer
3. 74153 = 4-to-1 multiplexer
4. 74152 = 8-to-1 multiplexer
5. 74150 = 16-to-1 multiplexer.

Multiplexers are also used in digital communications for signal processing. There are several types of multiplexing techniques used in digital communication systems, including:

1. Time Division Multiplexing (TDM): TDM is a technique where multiple signals are transmitted one after the other over a single channel. Each signal is assigned a time slot or time interval, and the channel is shared among the signals in a cyclical manner.
2. Frequency Division Multiplexing (FDM): FDM is a technique where multiple signals are transmitted simultaneously over a single channel by dividing the frequency spectrum of the channel into multiple frequency bands, each band carrying a different signal.
3. Wavelength Division Multiplexing (WDM): WDM is a technique where multiple optical signals are transmitted simultaneously over a single optical fiber by assigning each signal a different wavelength of light.

2)4:1 Multiplexer

A 4 to 1 MUX contains “FOUR” input lines and these are A,B,C and D two selected lines S_0 and S_1 and one output Z-line. Selected lines S_0 and S_1 select one of the four input lines to connect the outgoing line. The figure below shows a 4 to 1 MUX block diagram where, the multiplexer determines the input by the selected line. Following is the block diagram for 4x1 MUX. Configuration of selection lines(S_1, S_0) will determine the output of the MUX.



2.1) Truth table:

For input lines S_1 and S_0 , following is the truth table:

S_1	S_0	Out
0	0	D
0	1	C
1	0	B
1	1	A

Figure 1-Image Credits-allaboutelectronics

2.2) Expanded truth table:

Table 2: Truth table for (4x1) Multiplexer is as follow:

S0	S1	A	B	C	D	Y
0	0	A	0	0	0	A
0	1	0	B	0	0	B
1	0	0	0	C	0	C
1	1	0	0	0	D	D

3) Introduction to Transistors

Transistors are basic electronic components that play an important role in modern semiconductor and digital circuit technology.

They are essential for amplifying and controlling electric current.

In a typical field-effect transistor (FET), there are three main components: source, drain, and gate.

Source: Source is one of the two terminals of a transistor.

It serves as the point from which current enters the transistor.

Electrons move from the source to the drain when the transistor is in the “on” state.

Drain: Drain is the other terminal connection of the transistor and represents the point where current leaves the device.

The flow of electrons from source to drain is controlled by the gate.

Gate: Gate is the control element of the Transistor.

This is an insulating electrode used to determine whether the transistor is in the “on” or “off” state.

By applying a voltage to the gate, it is possible to regulate the flow of electrons between source and drain.

This modulation allows transistors to act as switches or amplifiers, making them the basis for digital logic gates and analog signal processing.

The combination of source, drain, and gate of a transistor forms the basis for the operation of countless electronic devices, from microprocessors and memory chips to amplifiers and integrated circuits.

Understanding the principles behind these components is essential for anyone working in the field of electronics and semiconductor technology.

3.1) NMOS Transistor

An NMOS (N-channel Metal-Oxide-Semiconductor) transistor is a type of field-effect transistor that is widely used in integrated circuit technology. NMOS transistors are fundamental building blocks in digital electronics and play a critical role in the creation of digital logic gates, microprocessors, and memory devices.

In an NMOS transistor, the key components are:

- 1) N-channel: The "N" in NMOS stands for the type of semiconductor material used in the transistor's channel. NMOS transistors use n-type semiconductor material, which is rich in electrons. When a voltage is applied to the gate terminal, it creates an electric field that allows the flow of electrons from the source to the drain, effectively turning the transistor "on."
- 2) Metal-Oxide-Semiconductor (MOS) structure: The NMOS transistor consists of a metal gate electrode separated from the semiconductor material by a thin insulating layer, typically made of silicon dioxide. This configuration enables precise control of the transistor's conductivity by

varying the voltage applied to the gate.

NMOS transistors are often used as the primary switching elements in complementary metal-oxide-semiconductor (CMOS) technology, working in conjunction with PMOS (P-channel MOS) transistors. The combination of NMOS and PMOS transistors allows for the design of energy-efficient digital circuits, making NMOS transistors a fundamental component in modern semiconductor devices. Understanding the operation and characteristics of NMOS transistors is essential for electronics engineers and designers, as they form the basis for creating the binary logic and memory elements that power our computers and electronic devices.

3.2) PMOS Transistor

A PMOS (P-channel Metal-Oxide-Semiconductor) transistor is a vital component in semiconductor technology and integrated circuits, serving as a complementary counterpart to the NMOS (N-channel Metal-Oxide-Semiconductor) transistor. PMOS transistors are used extensively in digital electronics and are instrumental in creating digital logic gates, memory devices, and microprocessors.

Key features of a PMOS transistor include:

- 1) P-channel: The "P" in PMOS signifies the type of semiconductor material used in the transistor's channel. PMOS transistors use p-type semiconductor material, which is deficient in electrons but has an abundance of "holes." When a voltage is applied to the gate terminal, it creates an electric field that allows the flow of "holes" from the source to the drain, effectively turning the transistor "on."
- 2) Metal-Oxide-Semiconductor (MOS) structure: PMOS transistors, like NMOS transistors, feature a metal gate electrode separated from the semiconductor material by an insulating layer, typically composed of silicon dioxide. This configuration enables precise control of the transistor's conductivity by varying the voltage applied to the gate.

PMOS transistors are commonly used in conjunction with NMOS transistors in complementary metal-oxide-semiconductor (CMOS) technology. The combination of PMOS and NMOS transistors allows for the design of energy-efficient digital circuits, making PMOS transistors a fundamental component in modern semiconductor devices. Understanding the operation and characteristics of PMOS transistors is essential for electronics engineers and designers, as they are crucial for creating binary logic gates and memory elements that underlie the operation of computers and electronic devices.

3.3) CMOS Transistor

A CMOS (Complementary Metal-Oxide-Semiconductor) transistor is a fundamental building block in modern integrated circuit technology. It is not a single transistor but a combination of both PMOS and NMOS transistors working together in a complementary fashion. This complementary operation is what gives CMOS its name.

Key features and characteristics of CMOS transistors include:

- 1) Complementary Pair: CMOS technology utilizes both PMOS and NMOS transistors on the same chip. PMOS transistors use p-type semiconductor material (electron deficiencies), while NMOS transistors use n-type semiconductor material (excess electrons). When combined, they create a complementary pair that can perform logical operations efficiently.
- 2) Low Power Consumption: One of the primary advantages of CMOS technology is its low power consumption. When a CMOS gate is not actively switching, it consumes minimal power, as one type of transistor is off (non-conducting) while the other is on (conducting). This makes CMOS ideal for battery-operated devices and energy-efficient circuits.
- 3) Digital Logic Gates: CMOS transistors are the building blocks of digital logic gates, such as AND, OR, NOT, and XOR gates. These gates are used to perform various logical operations in digital

circuits, enabling the creation of complex digital systems like microprocessors, memory chips, and integrated circuits.

- 4) Noise Immunity: CMOS logic gates exhibit excellent noise immunity. Since the PMOS and NMOS transistors are complementary, one is in an off state while the other is on. This property helps reduce the susceptibility to noise, making CMOS circuits highly reliable.
- 5) Voltage Compatibility: CMOS transistors can operate over a wide range of supply voltages, which makes them versatile and suitable for various applications.

CMOS technology is the foundation of modern semiconductor devices and digital electronics due to its low power consumption, high noise immunity, and versatility in implementing complex digital logic functions. It has revolutionized the electronics industry and is the basis for the design of many electronic devices we use today.

4) Question and answers:

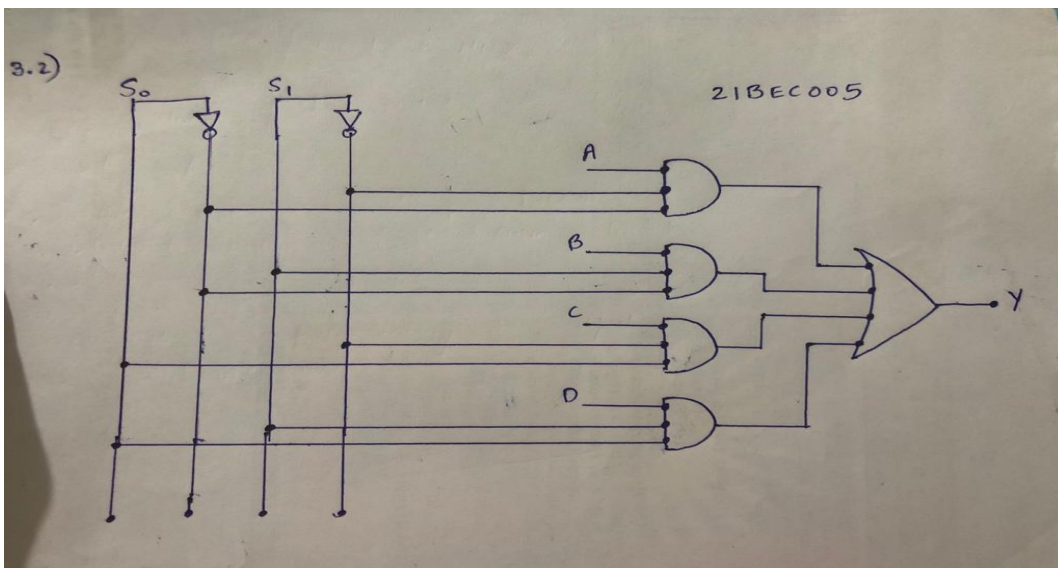
4.1) Find out the optimized Boolean equation for 4:1 MUX?

Ans. From the truth table provided in introduction, we can derive the Boolean equation:

$$Y = S_1' \cdot S_0' \cdot A + S_1' \cdot S_0 \cdot B + S_1 \cdot S_0' \cdot C + S_1 \cdot S_0 \cdot D$$

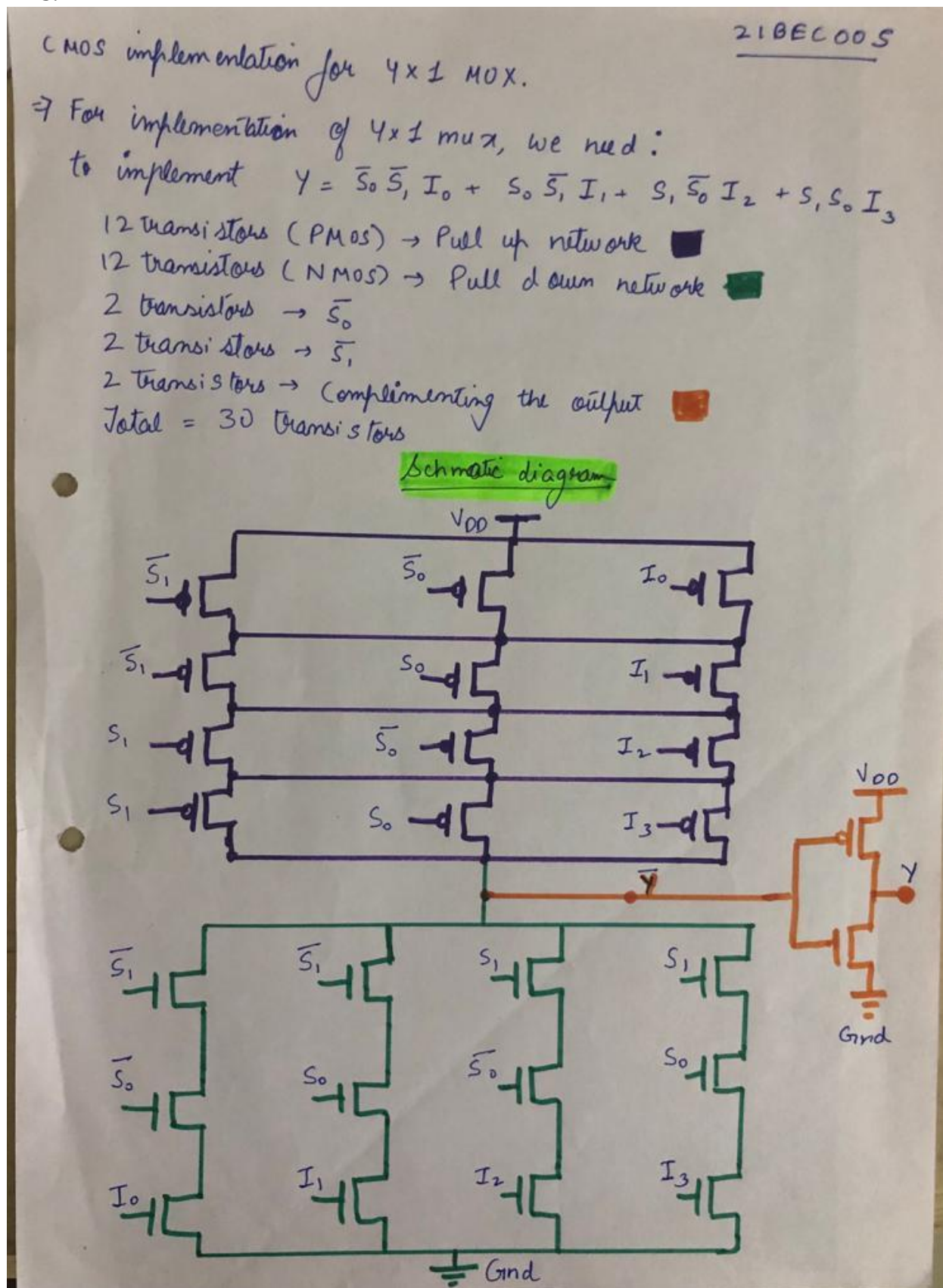
4.2) Draw the optimized gate level circuit diagram.

Ans. Following is the logical circuit for 4:1 MUX:



4.3) Draw the transistor level schematic for CMOS implementation.'

Ans.



4.4) Draw Transistor level schematic using NMOS gates.

Ans.

21BEC005

Designing 4x1 mux using Pass Transistor logic

$$Y = \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$$

$$= \bar{S}_1 (\bar{S}_0 I_0 + S_0 I_1) + S_1 (\bar{S}_0 I_2 + S_0 I_3)$$

Using NMOS transistor:

Input = 0 → NMOS OFF
Input = 1 → NMOS ON

4.5) Transistor level schematic using CMOS pass gate

Ans.

21BEC005

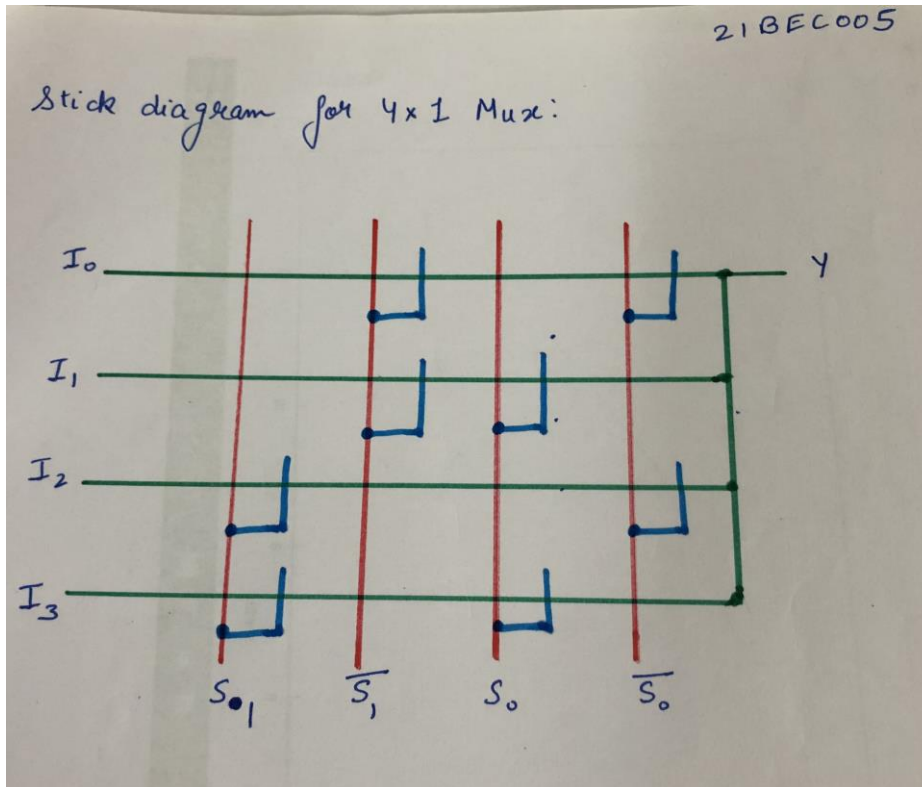
* Designing 4x1 Mux using Pass Transistor logic

NMOS - PMOS:

$$Y = \bar{S}_1 \bar{S}_0 I_0 + \bar{S}_1 S_0 I_1 + S_1 \bar{S}_0 I_2 + S_1 S_0 I_3$$

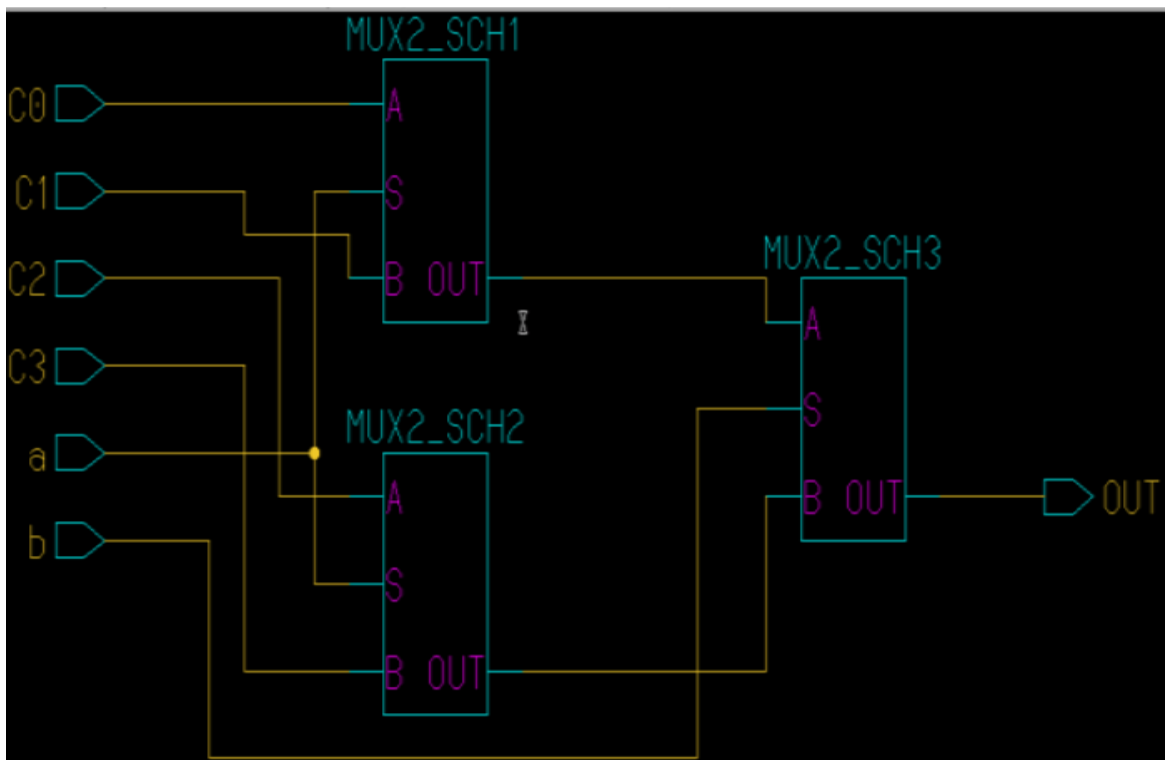
4.6) Draw stick diagram for 4:1 MUX using proper colour code.

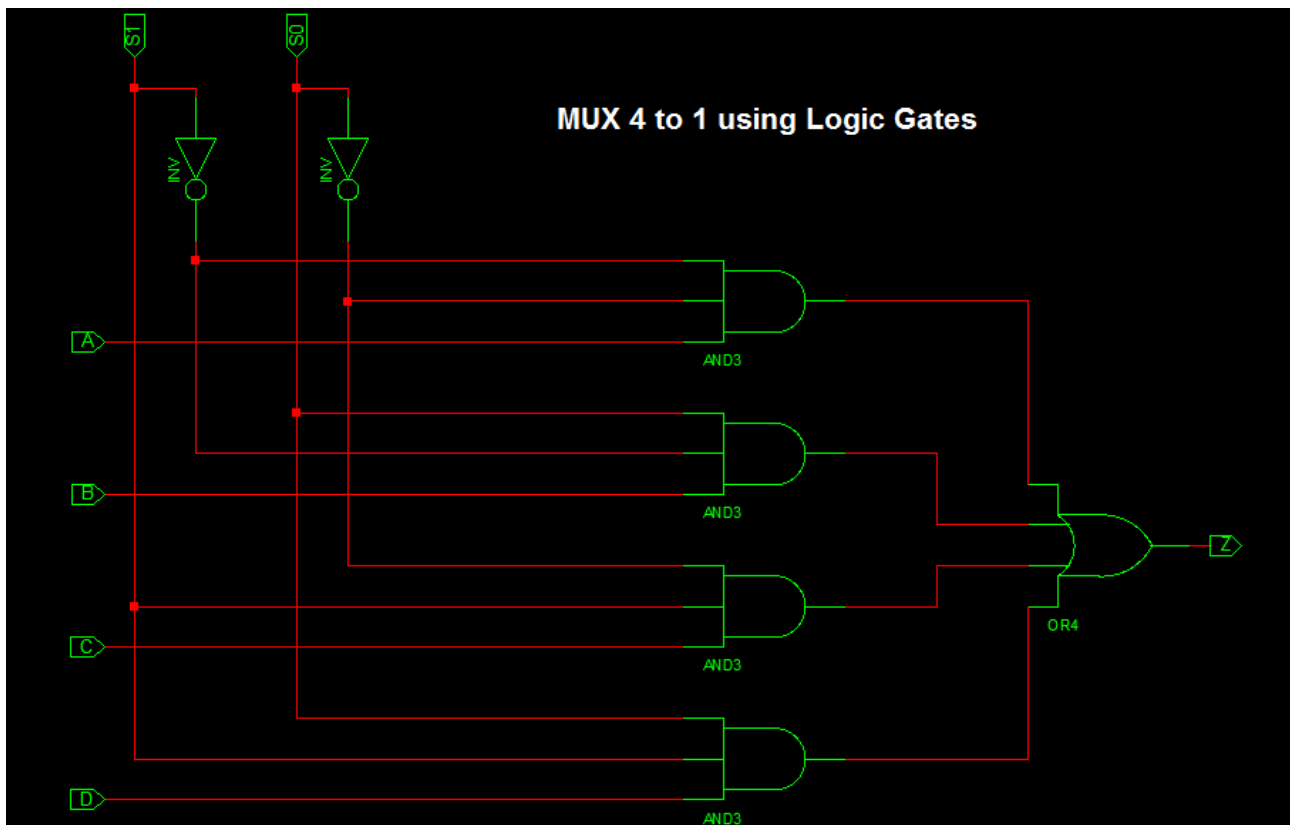
Ans.



4.7) implement 4:1 MUX using DSCH

Ans.

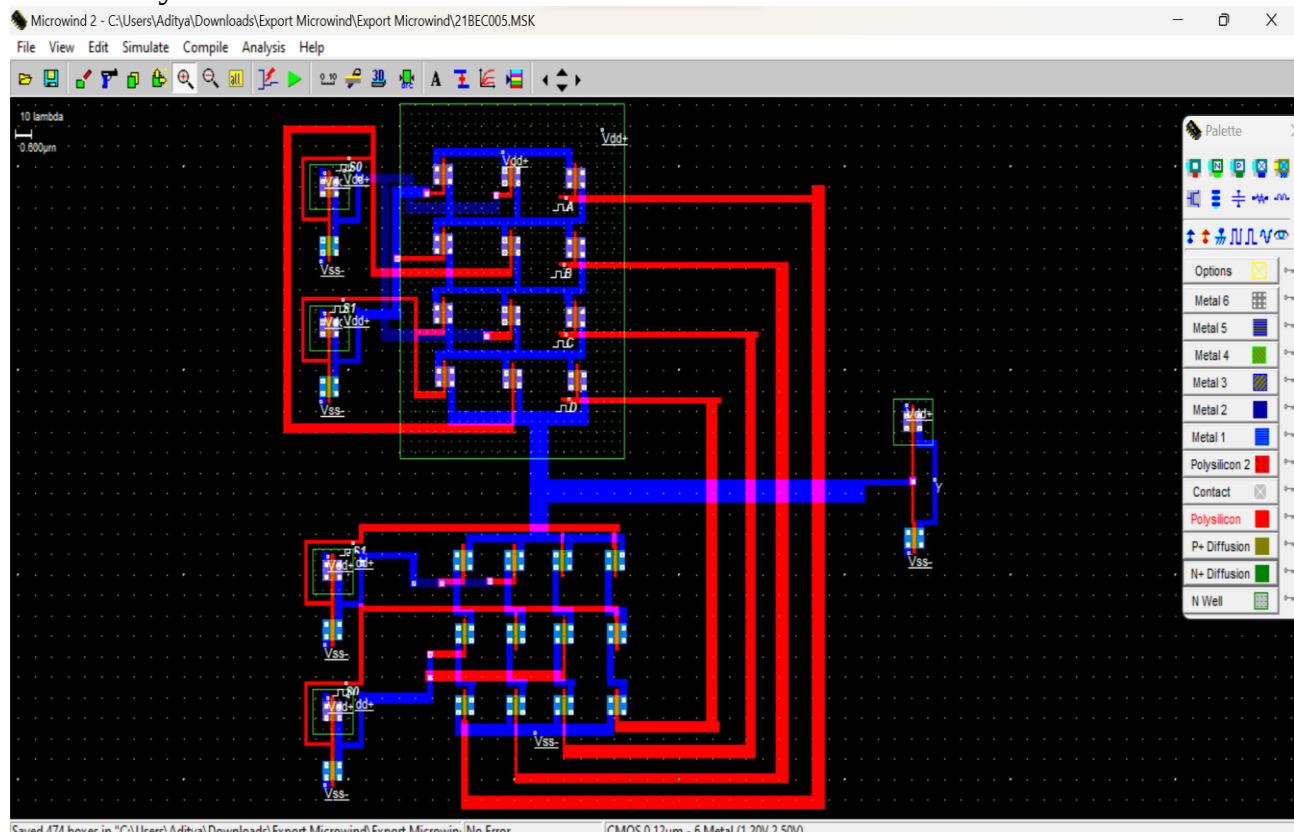




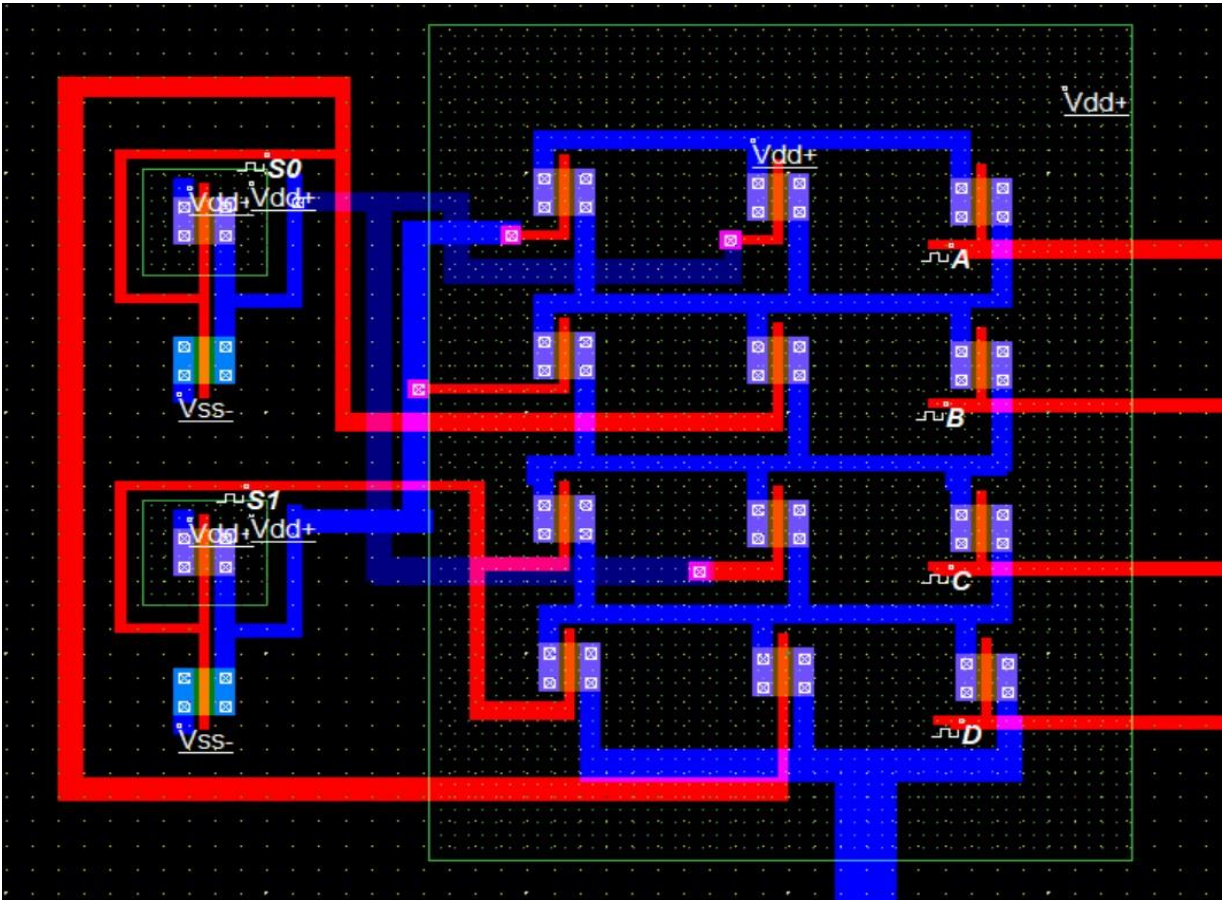
4.8) Layout for 4:1 MUX using Microwind.

Ans. First we are implementing 4x1 MUX using CMOS implementation:

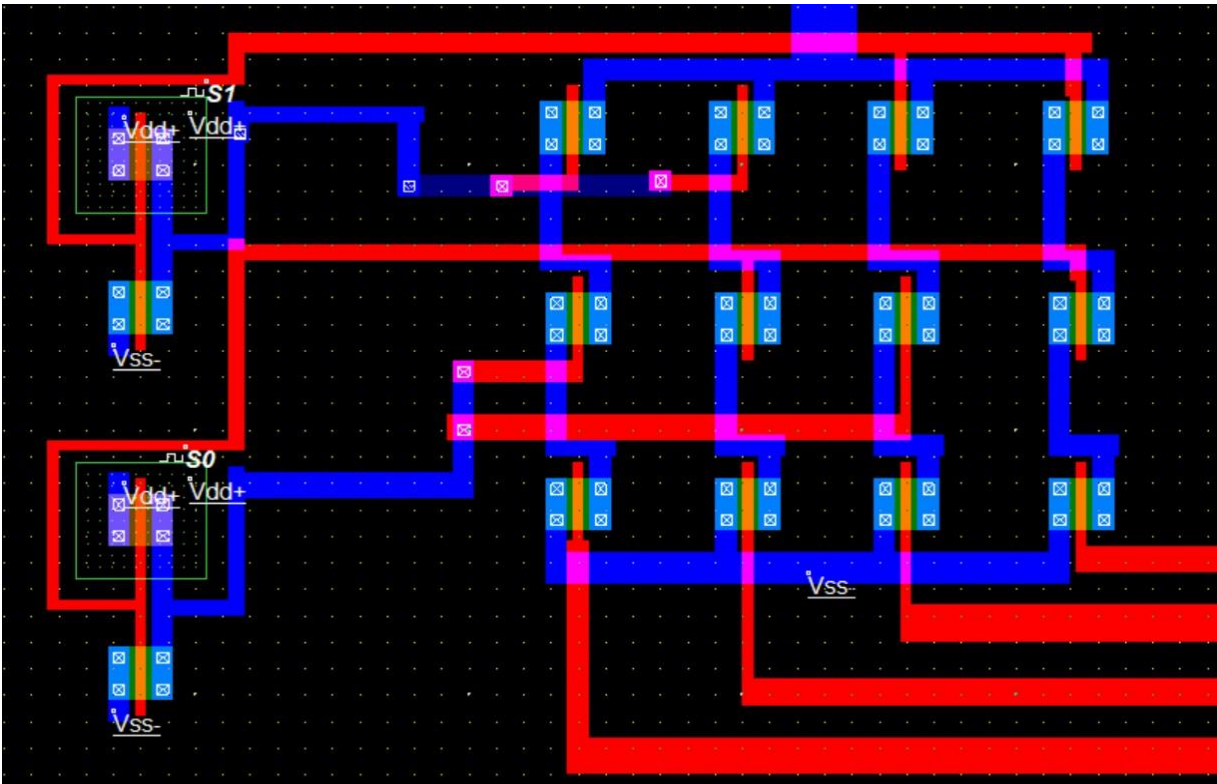
Overall layout:



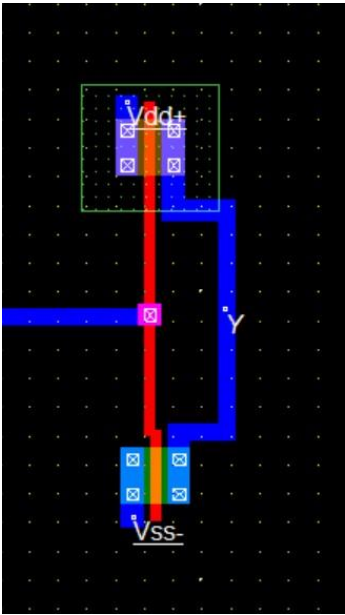
Pull up network:



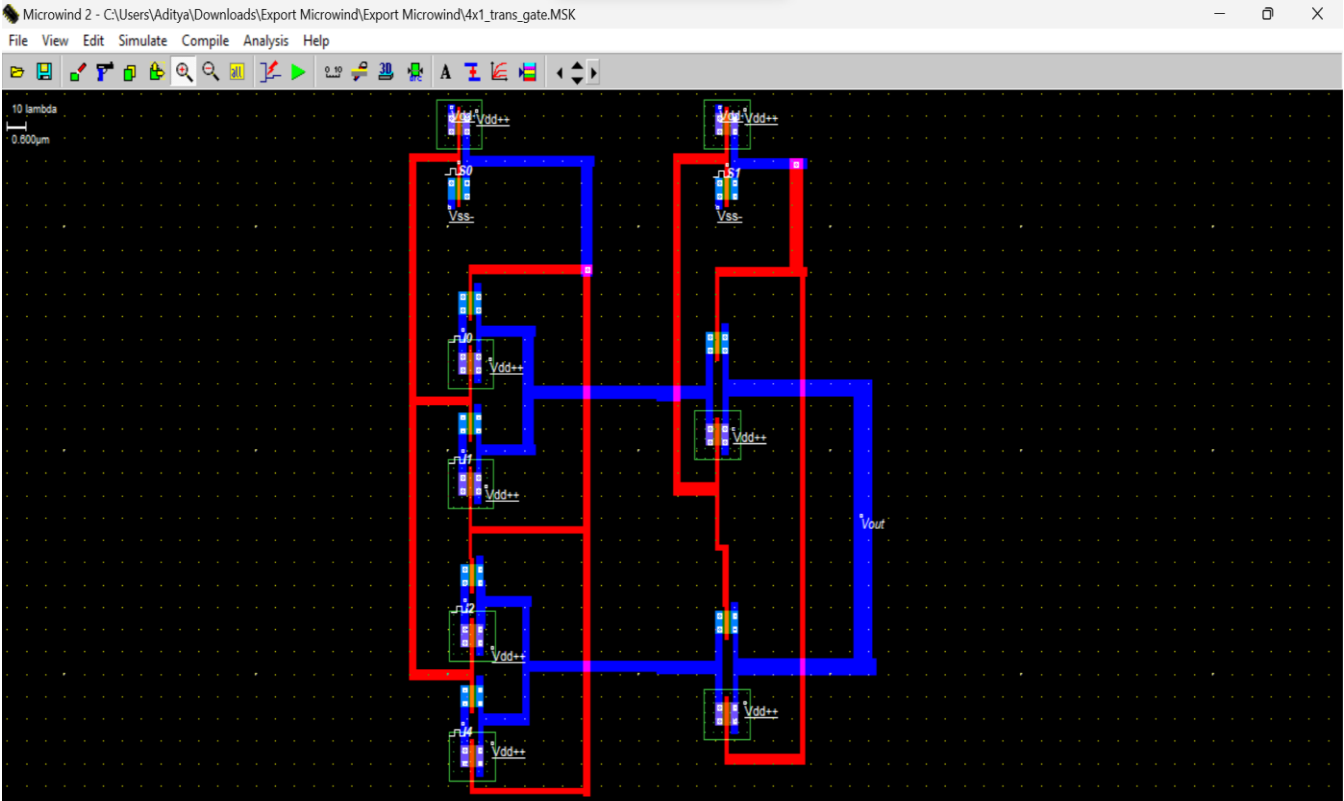
Pull down network:



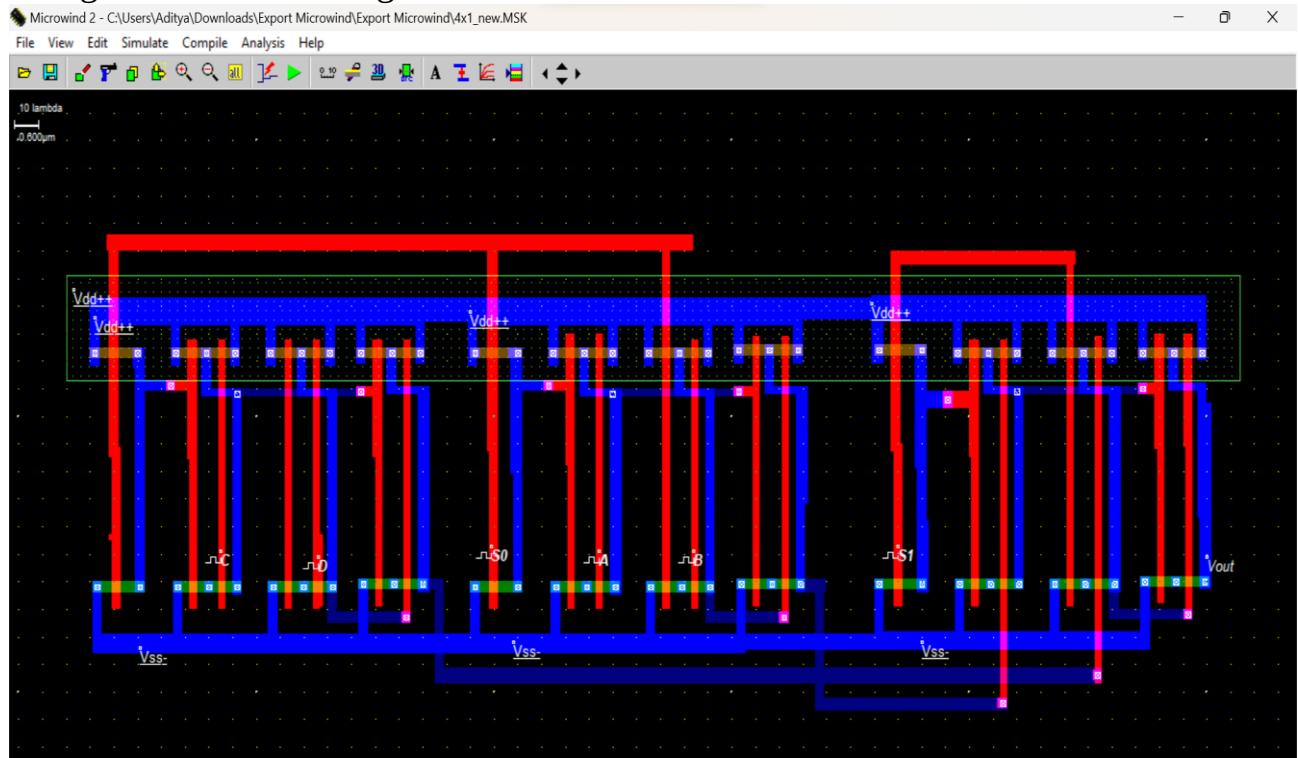
Output invertor:



Using CMOS transmission gate:

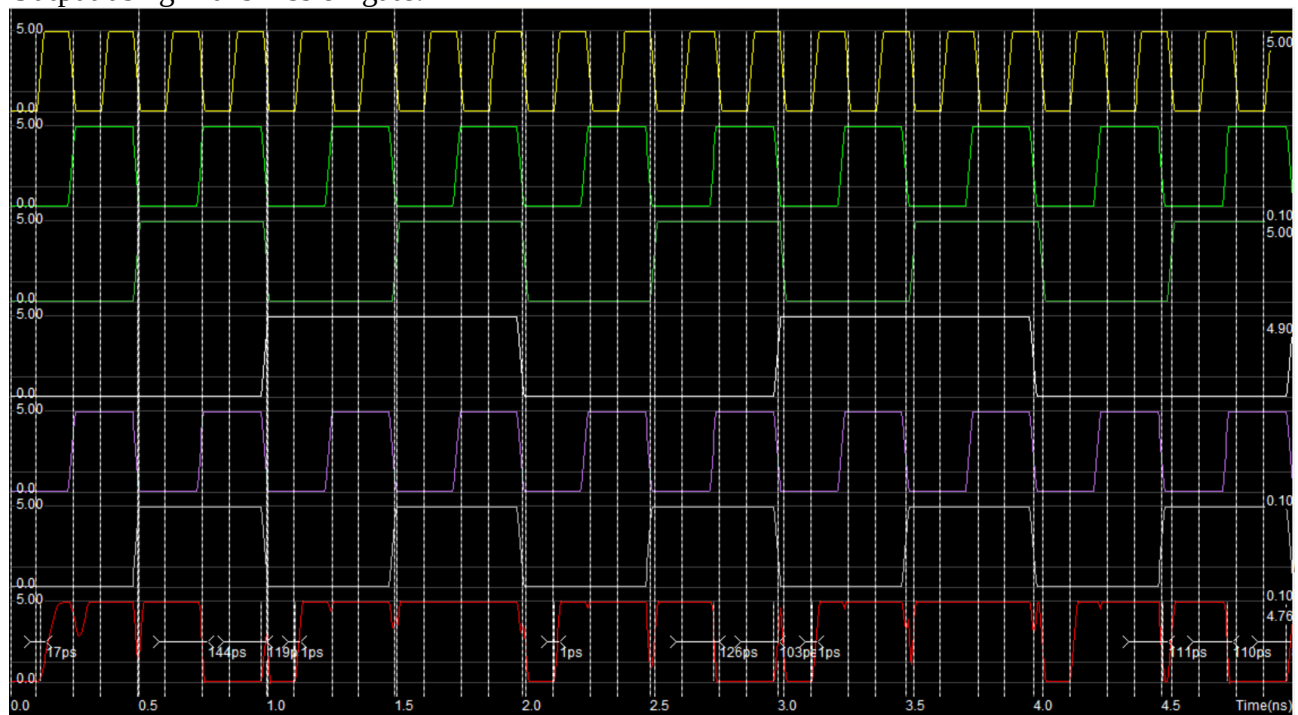


Using NOR and NAND gates:

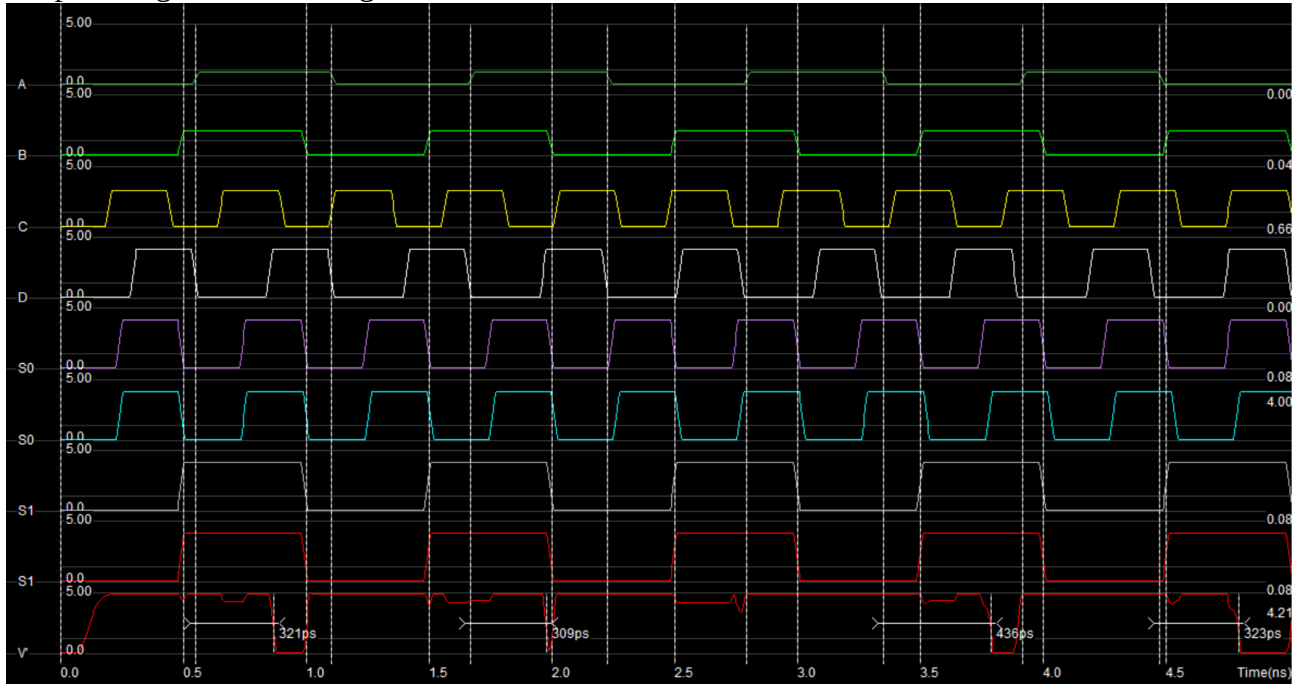


4.9) Simulation and observations:

Output using Transmission gate:



Output using NAND-NOR gate:



In the above simulation, I observed that mux using transmission gate implementation gives the best output upon comparison of output among CMOS implementation and fingers-NOR and NAND gate implementation.

CMOS implementation gives wrong output which is not expected.

Upon varying the value of R, width of driver, I received following values of Voh, Vol, Vil, Vih, Vth, Tphl and Tplh:

Technology (L)	Value of R (k)	W of Driver	V _{OH}	V _{OL}	V _{IL}	V _{IH}	V _{TH}	T _{PHL}	T _{PLH}
120nm	1	4λ	1.2	---	0.51	---	1.01	---	---
120nm	12	4λ	1.2	0.07	0.36	0.83	0.63	2ps	2ps
120nm	50	4λ	1.2	0.02	0.35	0.55	0.52	1ps	6ps
180nm	12	10λ	1.8	0.04	0.6	1.09	0.92	5ps	6ps
180nm	50	10λ	1.8	0.01	0.57	0.76	0.71	18ps	4ps
180nm	12	4λ	1.8	0.04	0.65	1.18	0.97	6ps	6ps
180nm	50	4λ	1.8	0.01	0.56	0.79	0.73	21ps	5ps
350nm	12	10λ	3.5	0.19	0.70	1.57	1.34	38ps	36ps
350nm	50	10λ	3.5	0.05	0.56	1.05	0.96	32ps	84ps
350nm	12	4λ	3.5	0.46	0.73	2.41	1.78	40ps	26ps
350nm	50	4λ	3.5	0.12	0.64	1.31	1.18	40ps	48ps

I also observed that CMOS implementation gives very weak output due to involvement of large number of transistors in the circuit. In the above designs, I have used following number of transistors:

Design Type	Number of PMOS	Number of NMOS	Total
CMOS	17	17	34
Transmission Gate	8	8	16
NOR_NAND	21	21	42

Upon comparison of outputs, I observed that increase in the number of transistors result in decrease in the efficiency of the output. Each transistor is contributing to an overall delay in the output. As zero rise time and fall time is not possible in the inputs, there exists a contradiction between selection line inputs(S0,S1) and inputs(I0,I1,I2,I3). The output obtained in this contradiction period is not desired. Still I have managed to minimize the rise time and fall time to 0.08ns and 0.09ns respectively. To deal with the aforementioned problem, we can implement the circuit using capacitors which can hold the output until the configuration of S1 and S0 changes. It will reduce the transition time from one output to another as output will not change during rise time or fall time. Capacitor must have a discharging path through which output will differ only upon transition. Another way to minimize the false output during contradiction time is to use an efficient threshold(Noise Margin) from which circuit can differentiate until what value of voltage, it has to treat as input low(IL) and input high(IH). I have observed that length and width of transistors are also affecting the output. As W/L ratio is being increased for transistors, output gets weak and does not match with the high level voltage(V_{dd}) of the input. During simulation, I observed that clock input(S0,S1) frequency must be feasible for input(I0,I1,I2,I3) frequency. Too high or too low frequency of clock inputs will result in undesired output.

****END****